# AMAL RAJ BABU RAJ

**Senior Software Engineer | Admin, Service and Platform Engineering | AWS Certified Practitioner**
+1 2265073380 | amalrajbrr@gmail.com | https://www.linkedin.com/in/amalraj-offl/ | https://amalrajbr.github.io | Ontario, Canada

## PROFESSIONAL SUMMARY

Senior Software Engineer with 7+ years building scalable backend and platform systems in Python (Django, FastAPI) and Elixir, with full-stack delivery in React/TypeScript. Known for automating high-risk manual operations into auditable, reliable workflows, with measurable gains such as a 50x I/O speedup and 4x API performance improvement. Hands-on with AWS, Kubernetes progressive delivery (Argo Rollouts, canary, feature flags), enterprise security, and AI-powered tooling. AWS Certified AI Practitioner (perfect score).

## CORE TECHNICAL SKILLS

**Programming Languages:** Python (3.8 - 3.14, asyncio), TypeScript, JavaScript, Elixir, Erlang, Bash
**Web Frameworks:** Django Ninja, Django REST Framework, Pydantic, FastAPI, Flask, Phoenix 1.8.3+ , Erlang Chicago Boss
**Frontend:** React, TypeScript, Tailwind CSS, Vite
**Cloud and Container Platforms:** AWS (ECS, KMS, CloudWatch, Lambda, Bedrock), OpenShift (OCP), Docker Swarm, Docker, Kubernetes
**Deployment & Release Engineering:** Argo Rollouts, Canary and Blue-Green Deployments, Feature Flags, GitOps, CircleCI
**Databases:** PostgreSQL, MySQL, SQLite, Oracle, SQL Server, MongoDB, Redis, Mnesia
**Messaging & Async Processing:** Redis Queue (RQ) and RQ Scheduler, Kafka, RabbitMQ, asyncio concurrency
**Observability & Tracing:** Sentry, New Relic, Pydantic Logfire, Elastic
**AI & Automation:** LLM Integration, AI Agents, MCP Server Development, Prompt Engineering, PydanticAI, OpenAI API, Anthropic API, AI Policy and Governance
**Testing & Code Quality:** Pytest, Unit and Integration Testing, Code Coverage, uv, prek (pre-commit hooks), automated linting
**Tools:** VS Code, Cursor, GitHub Copilot, Postman, Insomnia, Jira, ServiceNow, Confluence, ClickUp, Slack
**Methodologies:** Agile, Scrum, Waterfall

## PROFESSIONAL EXPERIENCE

**Senior DevOps Developer | Royal Bank of Canada, Canada** *July 2026 - Present*

• Architected an end-to-end **automation platform** (Django, React/TypeScript, Redis) for daily FRA analyst operations, absorbing the workload previously handled by a full-time contractor; one engineer now runs it with a single upload and an EOD report.
• **Automated Archer ingestion**, column-rule filtering, re-validation**,** and JSOC log sanity checks, then split results into per-brand (Check Point, Palo Alto), per-policy batches, **with locking to block actions on problematic policies**.
• Engineered a window-restricted staging engine for firewall rule attestation, staging only in two 20-minute windows each hour, concurrency capped at 2-3 requests, RQ workers plus a RQ scheduler poller, and **retries with thresholds** and **backoff** that push batches through a **~40% first-pass success rate**. Added **manual status override** for unreliable poll results.
• Built an **analyst UI** with brand filters, policy search, selection and valid/invalid FRA stats, **5-second live updates**, and **downloadable run reports** and **manual staging batches** for contractors without system access.
• Delivered a **50x speedup** on Archer retrieval (100 FRA IDs: **~200 seconds down to ~4 seconds**) by batching FRA IDs into single API calls, parallelizing calls with an asyncio.Semaphore and event queue, and caching rule data so FRAs sharing a rule skip redundant lookups.
• Practiced progressive delivery on OpenShift (OCP) with canary deployments and Argo Rollouts, using feature flags and weighing trade-offs across rolling, blue-green, and canary strategies.
• Standardized repositories for **AI-assisted development with reusable skills and agent marketplace for plugins**, removed cross-repo dependencies, and **adopted uv and prek** to enforce linting, docstrings, and commit quality checks.
• Built a **step-strict AI agent** in RBC Assist with strong engineering standards incorporating negative prompts, few shot prompting, spot-checks, output validation, reducing analyst reliance on developers for report generation and verification.

**Software Engineer | Diversio, Canada** *January 2022 - June 2026 (4 years, 6 months)*

• Built **admin access aware sensitive customer data export desk** with server owned drafts, preview rows, column selection, filters with streamlined finalization and **read-only audit logs with filters persisted**
• Engineered **cross-env benchmark data** migration with **compare-before-import drift detection, advisory locks, RQ jobs** and **Slack notifications** converting manual high-risk operation to observable **auditable admin workflow**
• Boosted application performance by **4x** using **Django Ninja** vs **Django REST Framework**
• Led **Pulse Survey** project from conception to deployment within 4-week timeline
• Implemented enterprise-grade security including **Role-Based Access Control (RBAC)** in REST APIs, per-tenant **enveloped encryption** using **Fernet keys** with **AWS KMS**, and **blind indexing** for secure rapid search capabilities
• Developed in-house **MCP Servers** using **PydanticAI** and **OpenAI/Anthropic LLMs**, enhancing AI-powered analysis and reporting capabilities for business intelligence
• Automated **SurveyMonkey** data parsing pipelines reducing manual processing time by **90%**
• Resolved critical performance bottlenecks using **Sentry** monitoring, improving system uptime by **25%**

• Implemented robust **webhooks** with **Slack, Eventbrite, Zoom Events, ClickUp,** and **SurveyMonkey**
• Designed comprehensive authentication mechanisms with **2FA**, **SSO**, **JWT**, and **Knox**
• Managed CircleCI deployment pipelines to **AWS EC2** / **ECS** with **CloudWatch**
• Led **training** and **mentorship** programs for 5+ junior developers / interns, improving team productivity

**Research Assistant | Conestoga SMART Center, Canada** *May 2021 – December 2021 (8 months)*
• Real-time image manipulation using **Raspberry Pi** and **DLP2000 EVM** with **Scrcpy** and **OpenCV**
• Optimized hardware performance by monitoring MCU temperature using **Bash** and **Python FastAPI**
• Developed **SSH** control interfaces for **embedded systems**, enabling remote device management

**Software Engineer | Netstratum Technologies Limited, India** *May 2020 – August 2020 (4 months)*
• Led development team of 4 members for **HRM product** using **Erlang** programming language
• Designed comprehensive data models using **Xmind** and conducted thorough **API testing** using **Insomnia**

**Junior Programmer | Odyssey Technologies Limited, India** *June 2018 – March 2020 (1 year, 9 months)*
• Developed secure cryptographic solutions for **Public Key Infrastructure** using **Erlang**
• Built **Certificate Authority**, **eKYC**, **Data Vault**, and banking applications serving **10,000+** users
• Implemented multiple authentication protocols including **X509**, **password**, and **Xorkee** auth systems
• Developed **python scripts** to execute cryptographic functions that were expensive in Erlang

## EDUCATION

**Advanced Diploma** in **Embedded Systems Development** | Grade: **90.32%** (High Distinction)
Conestoga College, Cambridge Campus, ON | September 2020 – December 2021
**Bachelor of Engineering** in **Electronics & Instrumentation** | CGPA: **8.09/10**
Jerusalem College of Engineering, India | June 2014 – June 2018

## PROFESSIONAL CERTIFICATIONS

• AWS Certified AI Practitioner
• AWS Certified Cloud Practitioner
• Docker Mastery with Kubernetes + Swarm
• Django Features and Libraries
• Building Web Applications in Django
• Web Application Technologies and Django
• Certified Python Programmer
• Learning C++
• Learning Git and GitHub
• **Introduction to Industry 4.0** and Industrial IoT (IIT Kharagpur)

## TECHNICAL EXPERTISE

• **Backend Development:** Django, Django Ninja and FastAPI with Pydantic, ASGI/WSGI tuning, middleware, async batching and caching (50x Archer retrieval speedup), React/TypeScript dashboards.
• **Functional Programming:** Elixir/Phoenix LiveView, Erlang Chicago Boss, OTP/GenServer, Mnesia.
• **Cloud & Infrastructure:** AWS, Docker, Swarm, Kubernetes, OpenShift. Canary deployments with Argo Rollouts, feature flags, CI/CD with CircleCI, and the trade-offs between canary, blue-green and rolling releases.
• **Database Management:** SQL and NoSQL, query optimization, indexing, normalization, data modeling.
• **AI & Automation:** MCP servers, AI agents with guided validation workflows, LLM integration, AI-first repository standards.
• **System Architecture:** Monolith and microservices, RQ queues and schedulers, Redis caching, window-restricted workflows with retry and backoff, audit-compliant status lifecycles.
• **Security & Encryption**: RBAC, Fernet encryption with AWS KMS, blind indexing, per-tenant data isolation, audit logging.

## KEY ACHIEVEMENTS

• **Performance Engineering: 50x Archer retrieval speedup** (100 FRA IDs: **~200 s to ~4 s**) using batching, asyncio.Semaphore and rule caching. Earlier, cut a critical process from 83 s to 3 s with Python threading.
• **Process Automation:** Built the **FRA automation platform** that automated a full-time contractor's manual staging workflow into a single-click, audit-compliant process run by one engineer
• **Academic Excellence: High Distinction** (90.32%) in Embedded Systems Development, **Best Paper Award** at an international engineering conference, Department Topper for 4 semesters, Student Representative at Conestoga College.